Inside our planet

As Earth cooled, most of it turned to solid rock. It formed layers, with the heaviest material at the center (the core), surrounded by a deep, rocky mantle and a thin, brittle crust. Nuclear reactions inside the planet make the rocks much hotter than their normal melting point, but intense pressure keeps the rock solid, except for the outer core, which is liquid.

THE CORE

Most of the planet's weight is concentrated in the core, which is mostly iron mixed with nickel. The inner core is solid, but surrounded by a liquid outer core. Together, they form a metallic ball almost 4,350 miles (7,000 km) across—roughly the size of Mars. Some meteorites are made of these same metals, and may be fragments of shattered planet cores.

▸ IRON METEORITE
This meteorite looks rusty because, like Earth's core, it contains large amounts of iron. It probably came from the core of a planet destroyed long ago.

THE MANTLE

The core is surrounded by a mass of solid rock called the mantle. It is rich in iron and very heavy, but not as heavy as the metallic core. The mantle rock is extremely hot, and this makes it moldable, like modeling clay. The intense heat deep within the planet keeps the mantle moving very slowly, in currents that cause earthquakes and volcanic eruptions.